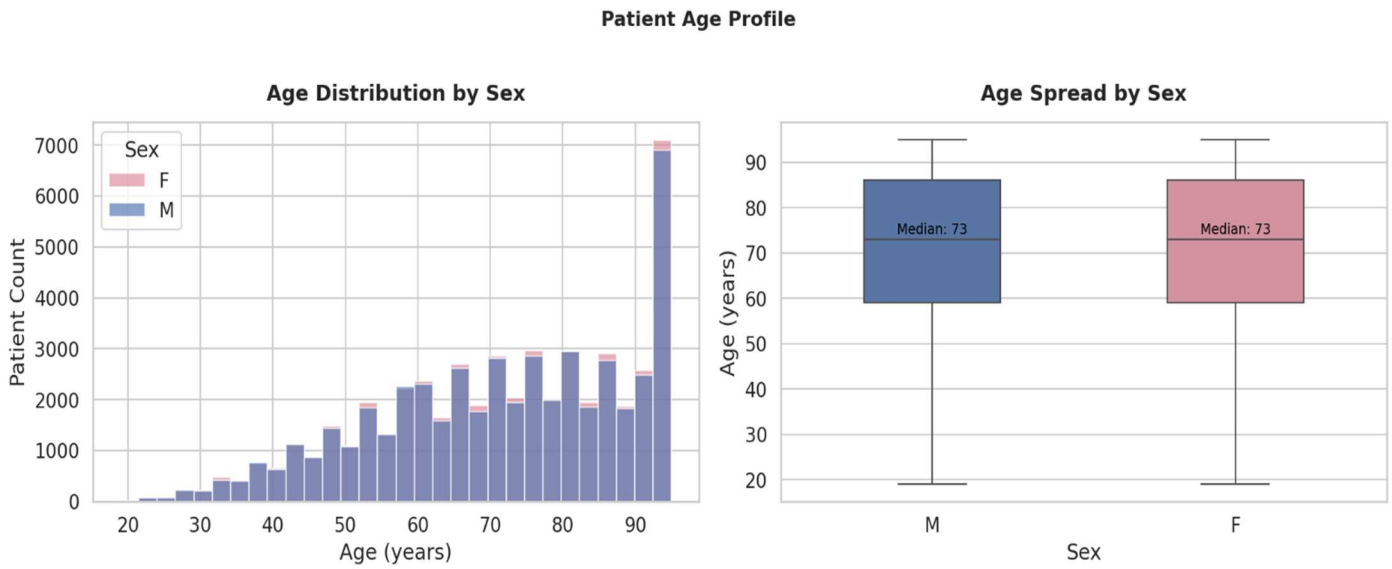
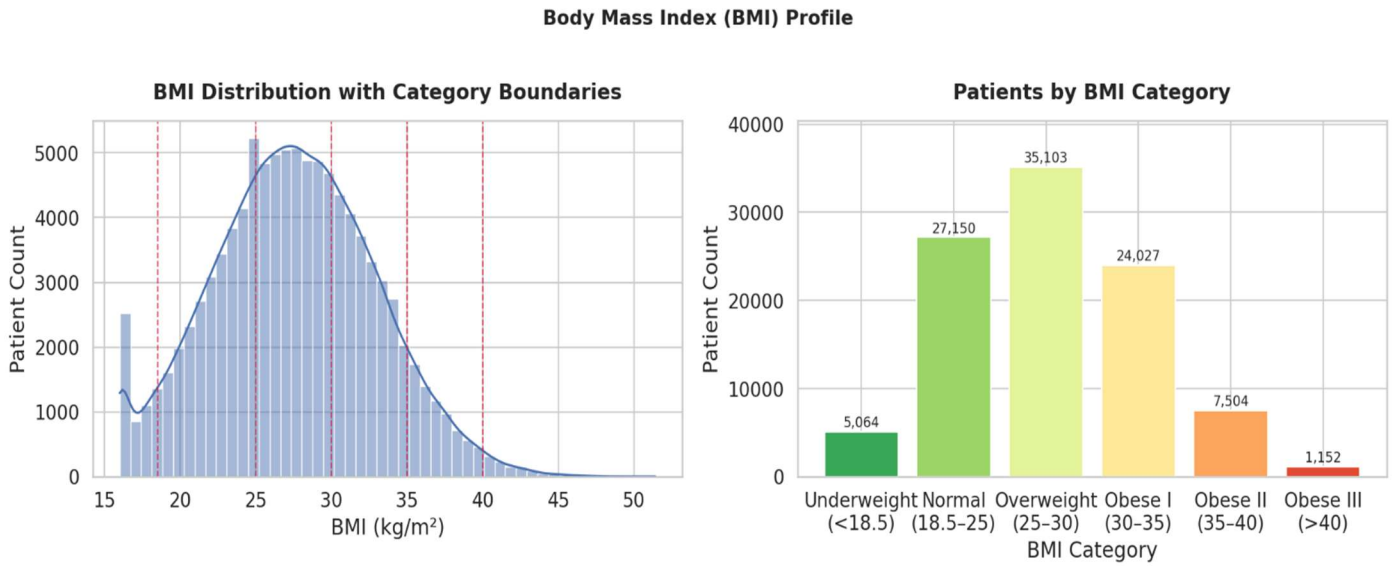
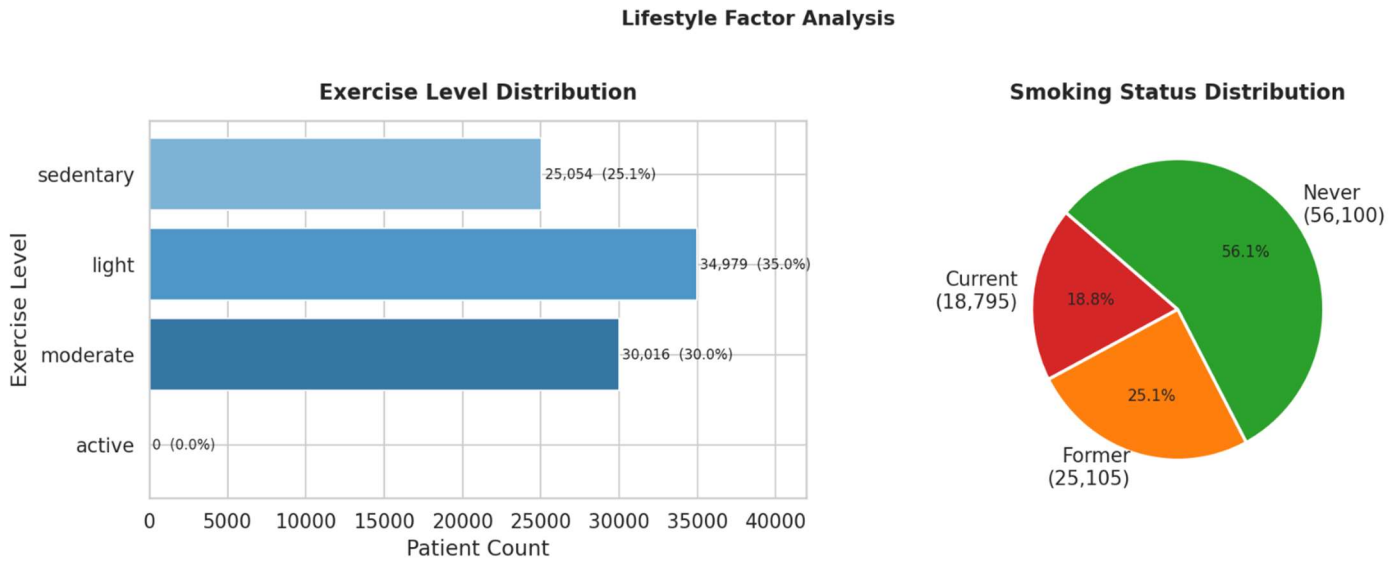
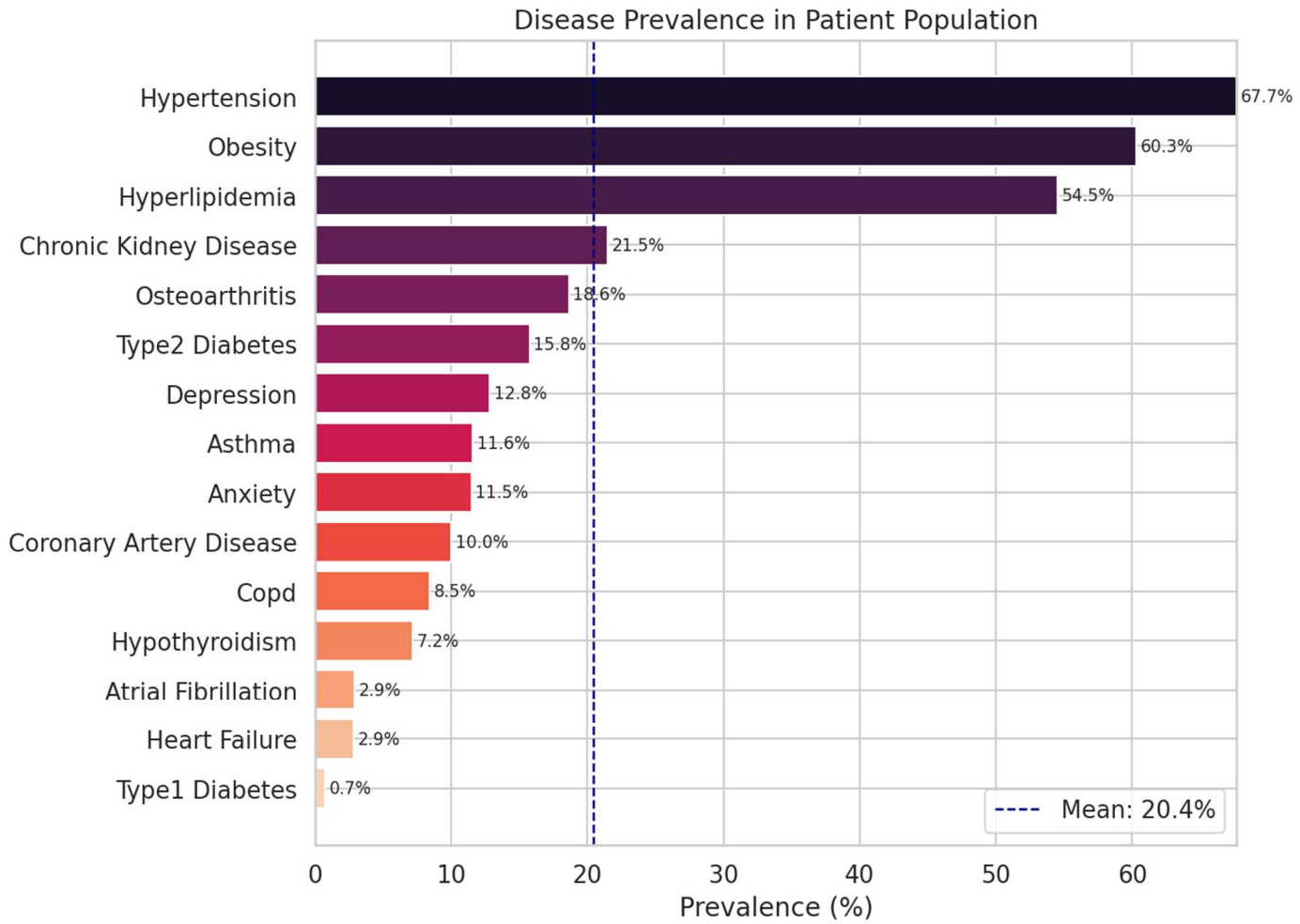
Clinical analysis was performed using five linked patient datasets from Kaggle, covering demographics, diagnoses, outcomes, laboratory tests, and medications:
https://www.kaggle.com/datasets/sergionefedov/patient-records-100k-patients-15-conditions/data

The workflow relied on **pandas, numpy, matplotlib, seaborn, plotly, and datetime** for data analysis and visualization.

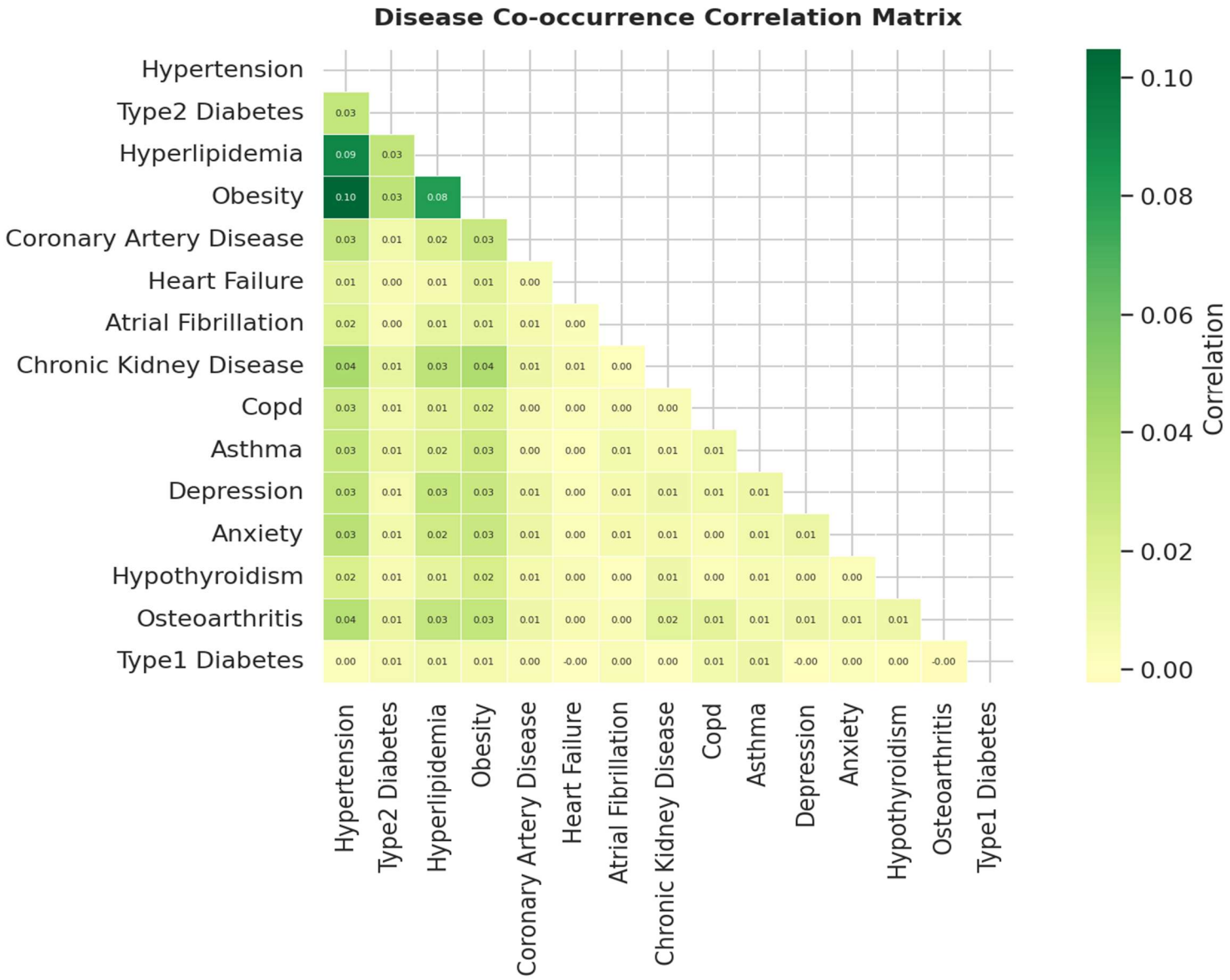


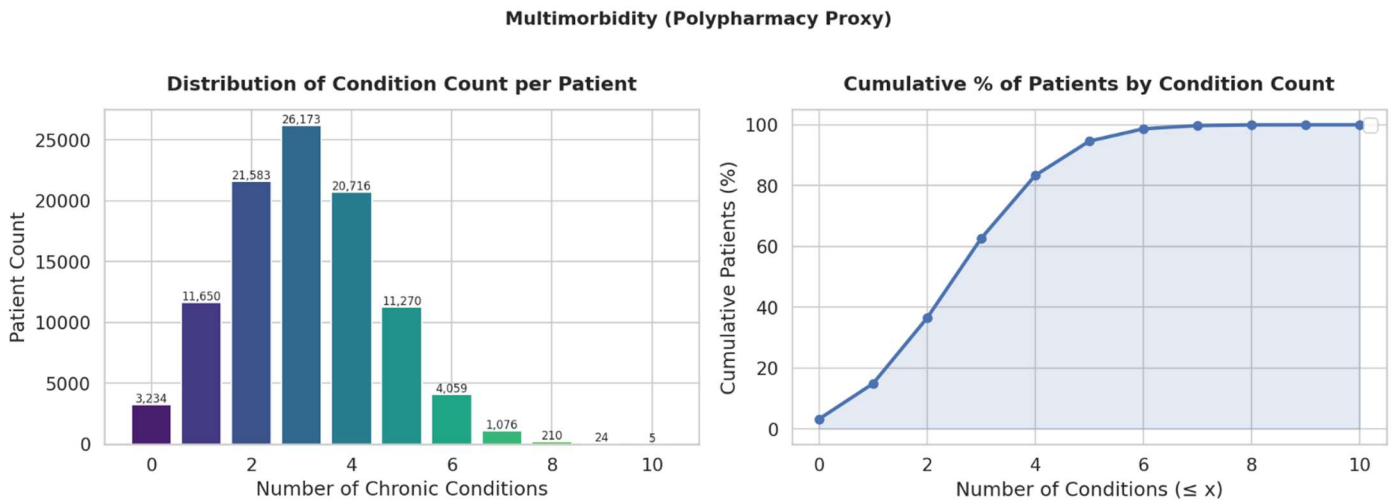




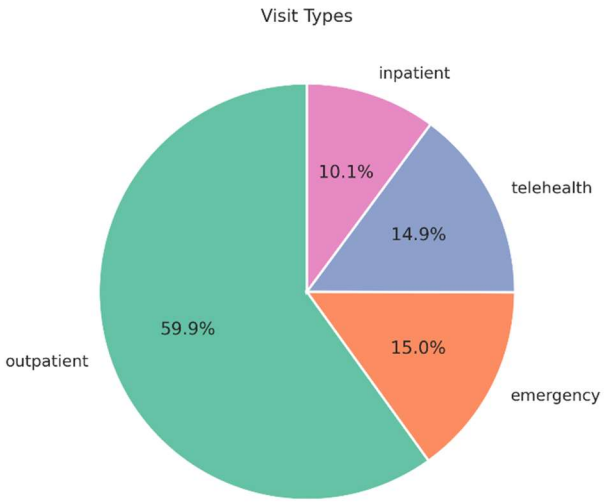

Disease Prevalence in Patient Population
Hypertension
Obesity
Hyperlipidemia
Chronic Kidney Disease
Osteoarthritis
Type2 Diabetes
Depression
Asthma
Anxiety
Coronary Artery Disease
Copd
Hypothyroidism
Atrial Fibrillation
Heart Failure
Type1 Diabetes
67.7%
60.3%
54.5%
21.5%
18.6%
15.8%
12.8%
11.6%
11.5%
10.0%
8.5%
7.2%
2.9%
2.9%
0.7%
Mean: 20.4%
0
10
20
30
40
50
60
Prevalence (%)

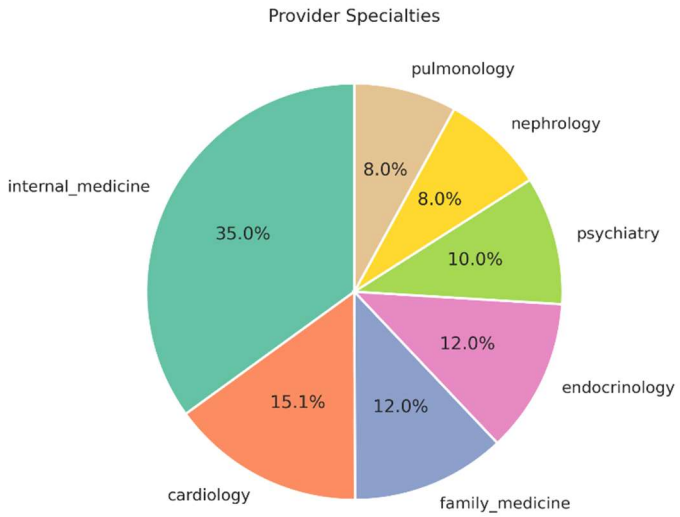
Disease Prevalence by Age Group (%)
Hypertension 27.3 37.1 49.8 63.9 81.6
Type2 Diabetes 7.4 8.7 11.5 14.9 19.0
Hyperlipidemia 20.4 29.6 40.1 51.1 66.0
Obesity 22.7 32.0 44.1 56.7 73.1
Coronary Artery Disease 3.7 5.6 7.3 9.3 12.1
Heart Failure 1.2 1.5 2.2 2.5 3.5
Atrial Fibrillation 1.7 1.6 2.3 2.7 3.5
Chronic Kidney Disease 7.5 11.5 16.3 19.9 26.0
Copd 3.7 4.5 6.3 8.1 10.1
Asthma 3.6 6.6 8.7 10.9 13.9
Depression 5.1 6.6 9.5 12.1 15.5
Anxiety 5.4 6.4 8.4 10.8 13.9
Hypothyroidism 2.6 4.0 5.4 6.8 8.6
Osteoarthritis 7.1 9.8 13.9 17.4 22.6
Type1 Diabetes 0.1 0.4 0.5 0.7 0.9
<30
30–44
45–59
60–74
75+
Age Group
Prevalence
80
70
60
50
40
30
20
10

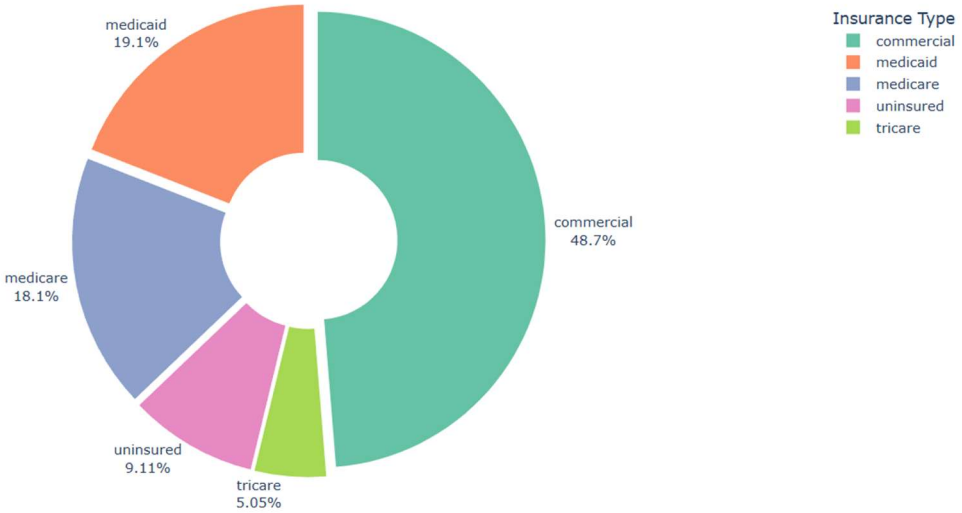


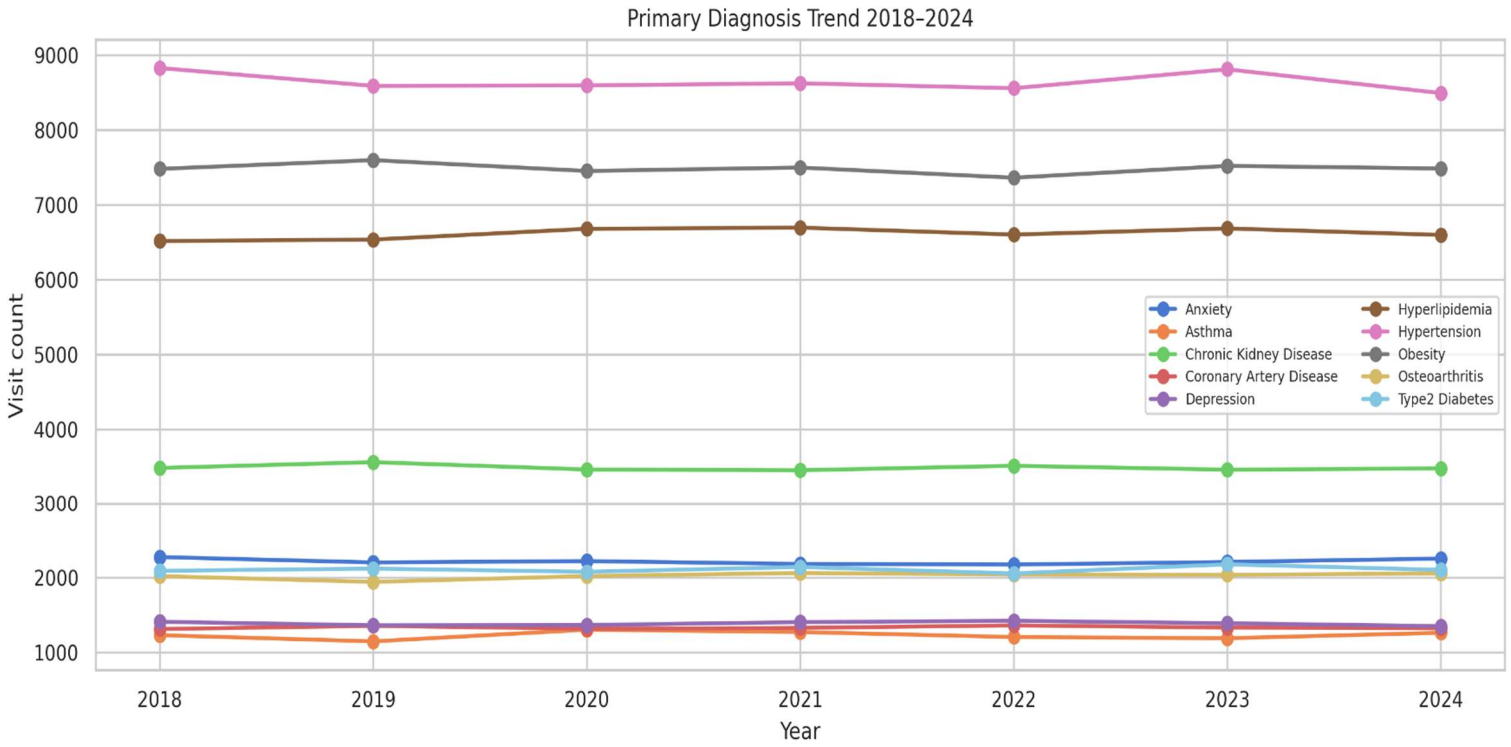



```
Multimorbidity Summary:
  Mean conditions per patient : 3.06
  Patients with 0 conditions  : 3,234
  Patients with 3+ conditions : 63,533
  Patients with 5+ conditions : 16,644
```

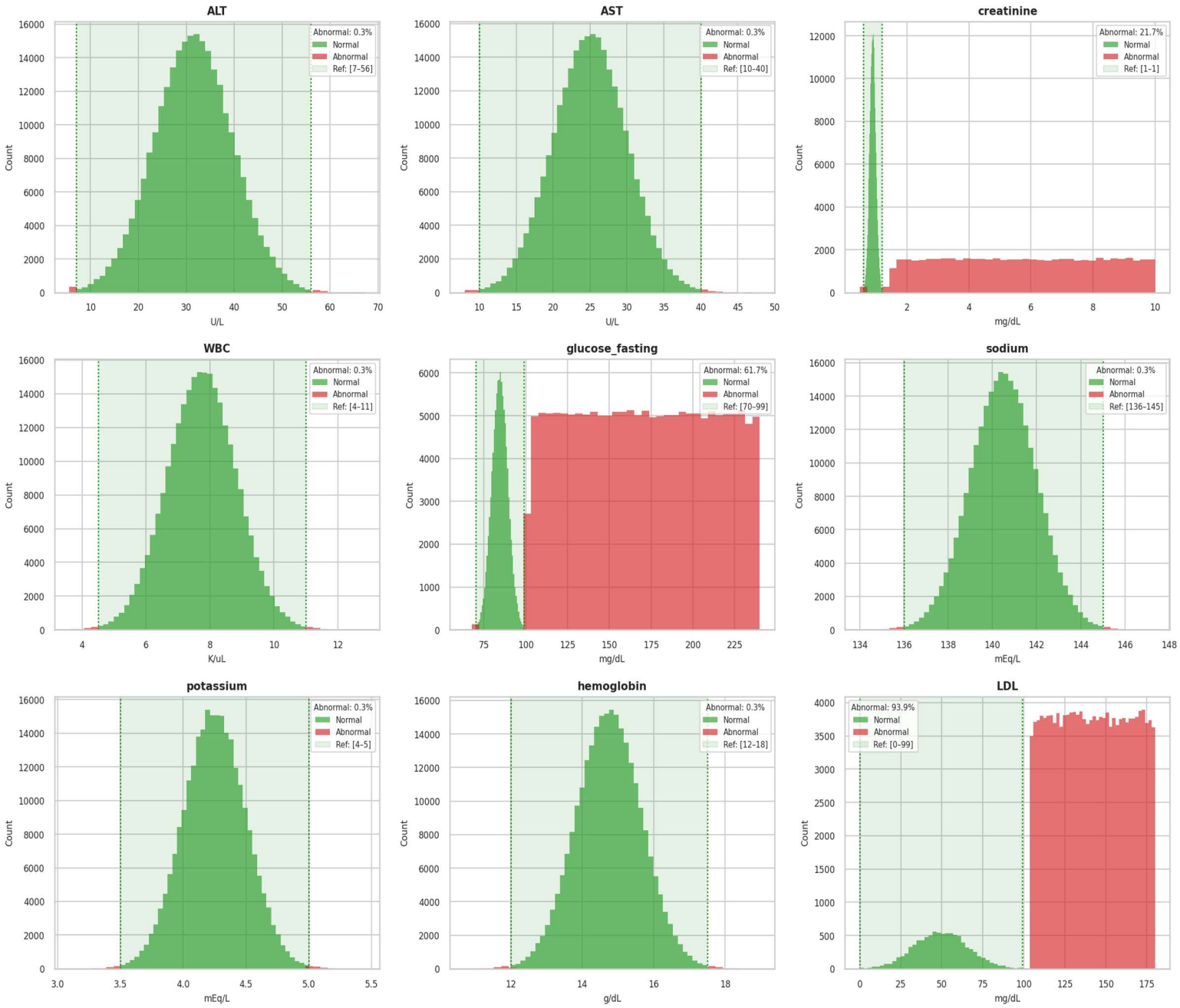

Visit Types
inpatient
10.1%
telehealth
14.9%
outpatient
59.9%
15.0%
emergency


Provider Specialties
pulmonology
nephrology
internal_medicine
8.0%
8.0%
35.0%
psychiatry
10.0%
12.0%
endocrinology
15.1%
12.0%
cardiology
family_medicine

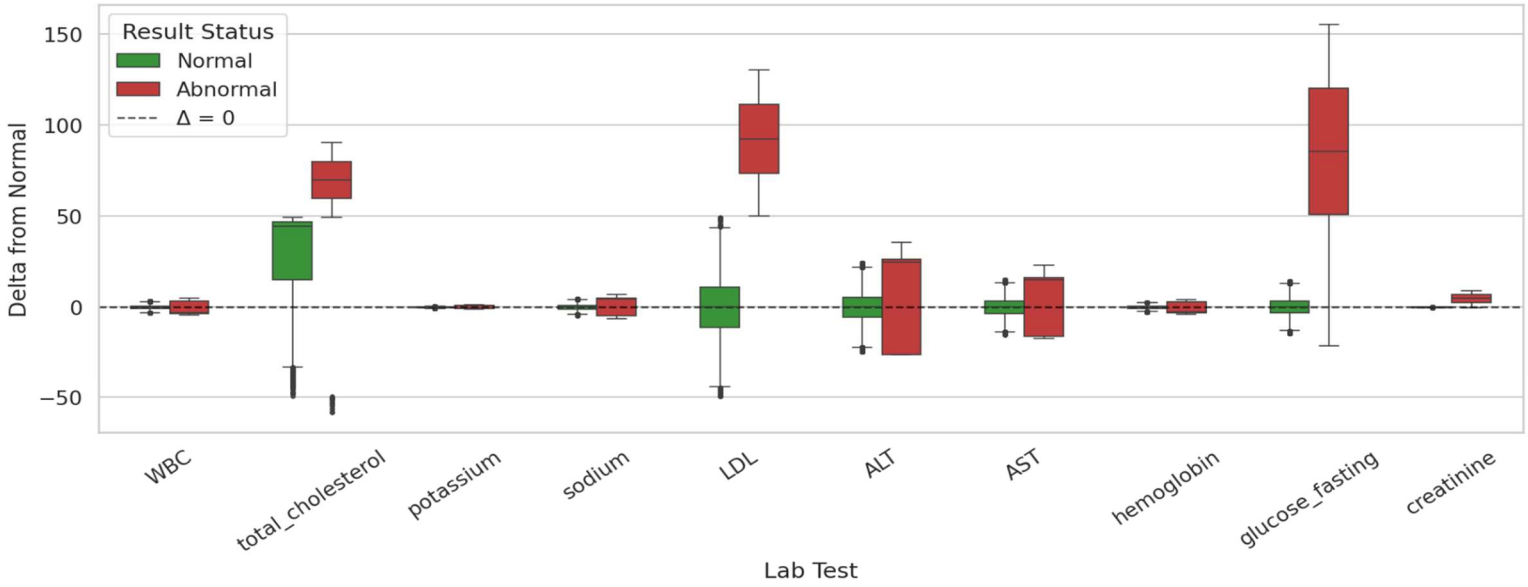

medicaid
19.1%
commercial
48.7%
medicare
18.1%
uninsured
9.11%
tricare
5.05%
Insurance Type
commercial
medicaid
medicare
uninsured
tricare

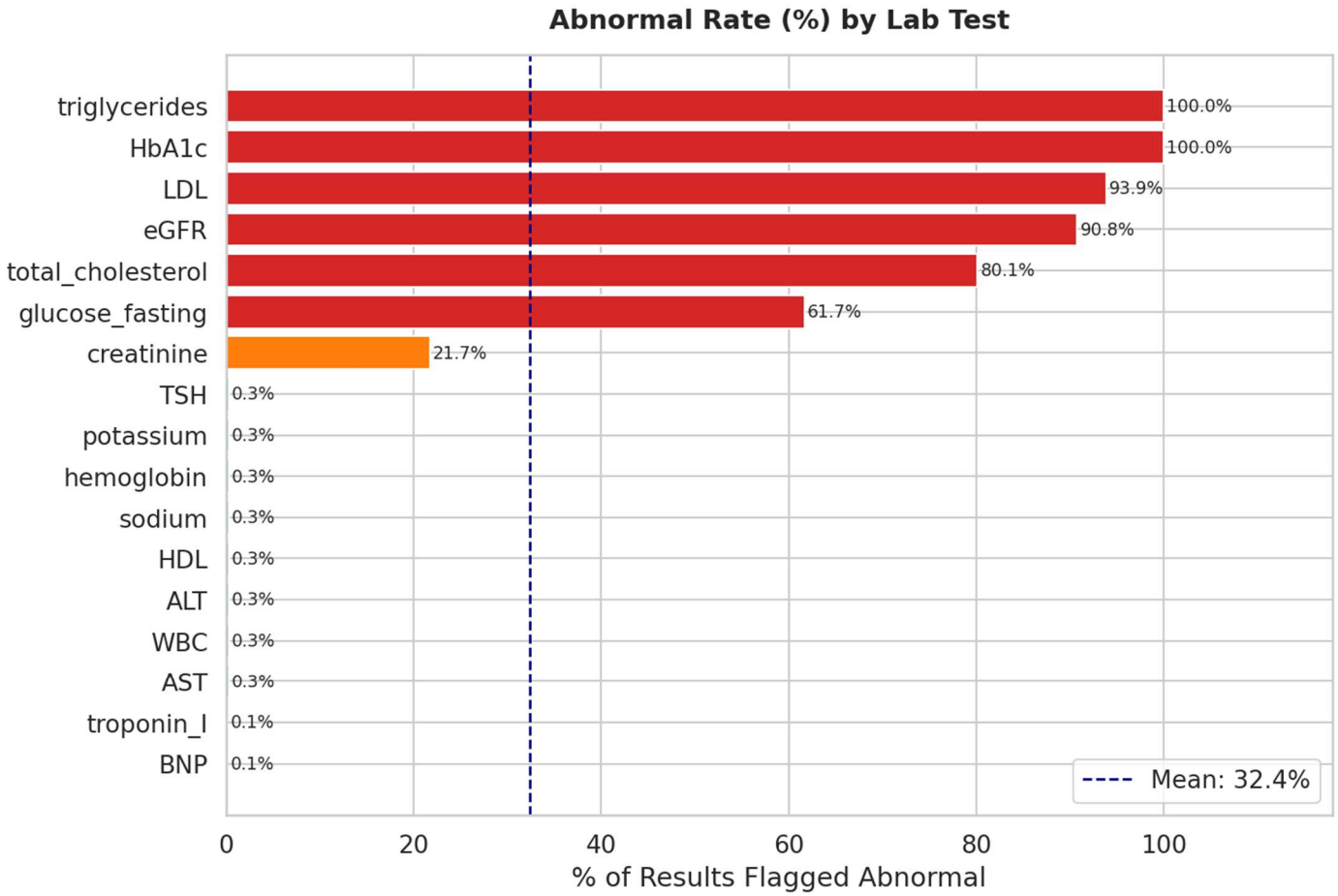

Primary Diagnosis Trend 2018–2024
Visit count
9000
8000
7000
6000
5000
4000
3000
2000
1000
2018
2019
2020
2021
2022
2023
2024
Year
Anxiety
Asthma
Chronic Kidney Disease
Coronary Artery Disease
Depression
Hyperlipidemia
Hypertension
Obesity
Osteoarthritis
Type2 Diabetes

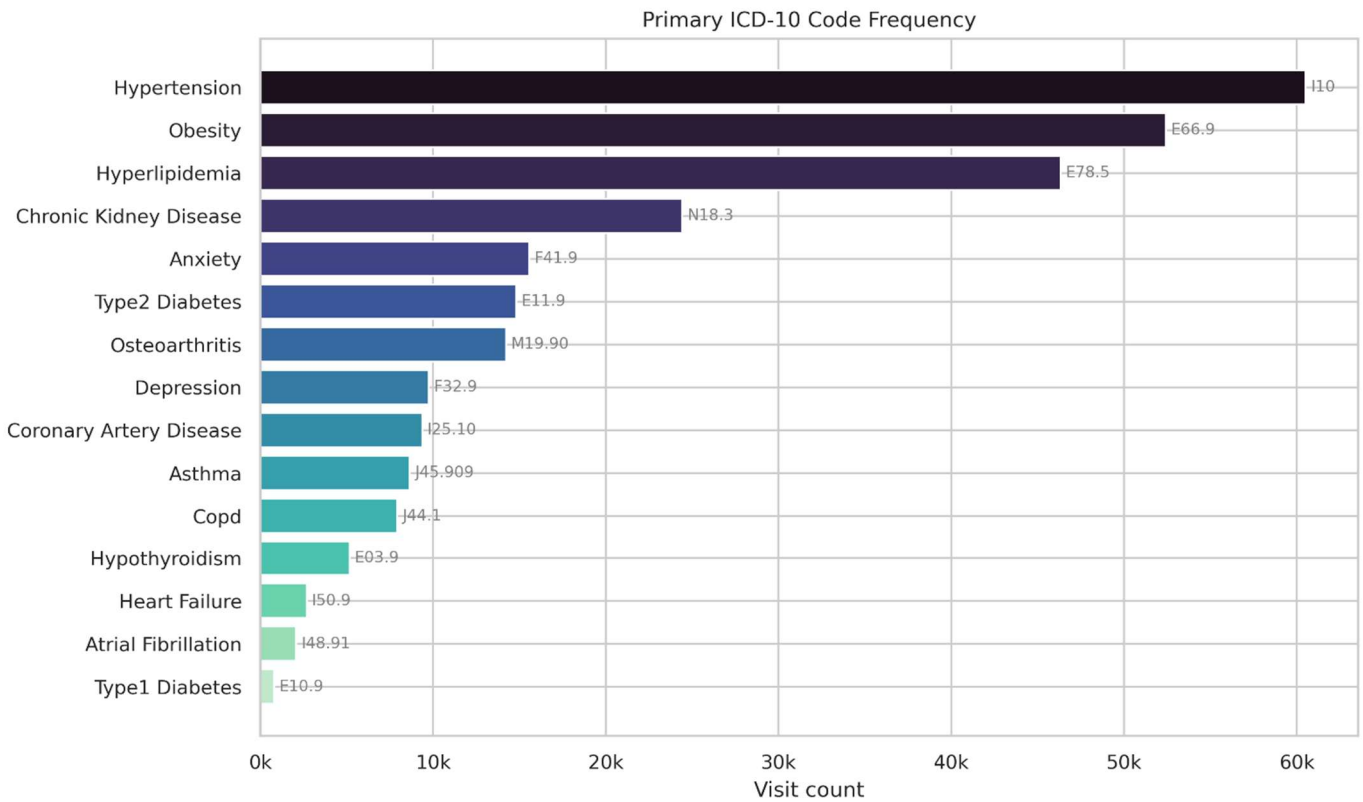
Lab Value Distributions — Normal vs Abnormal (Top 9 Tests)

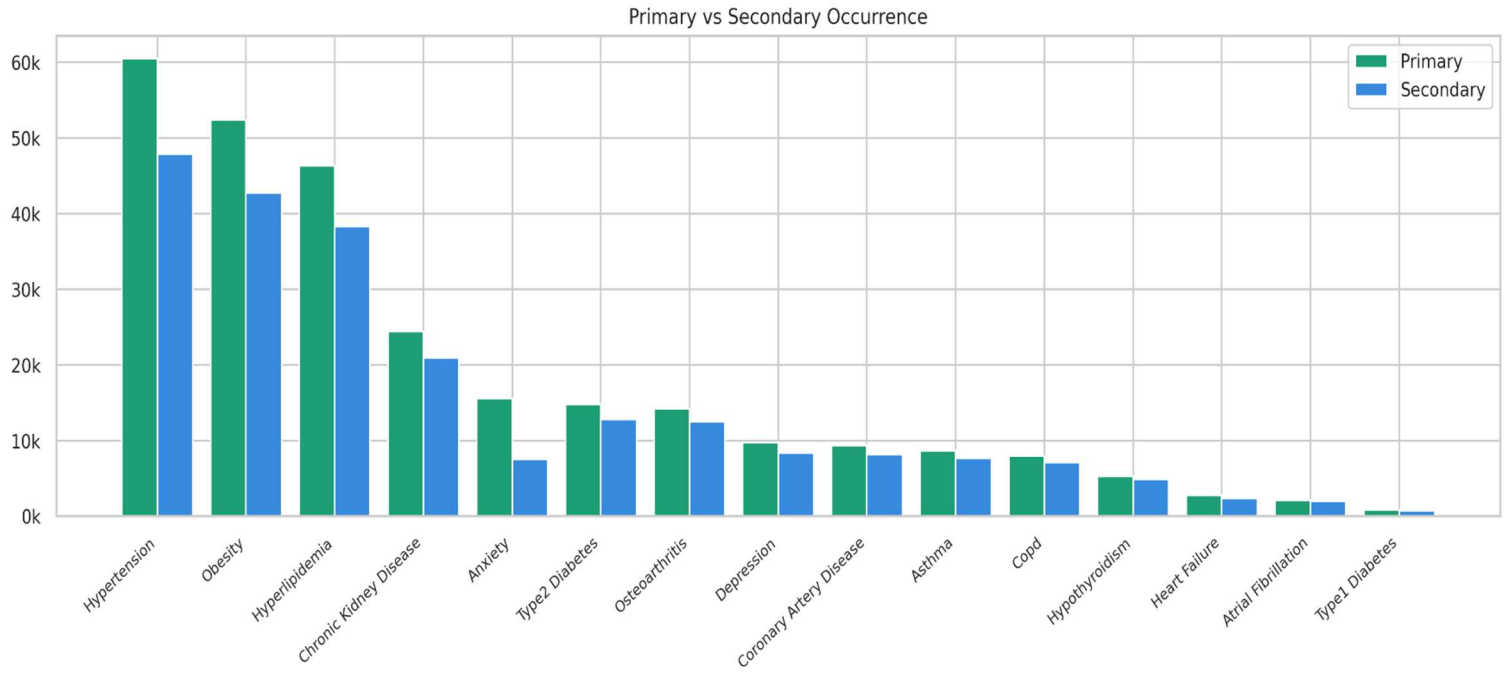


Delta from Normal by Lab Test (Normal vs Abnormal Results)




Abnormal Rate (%) by Lab Test
triglycerides
HbA1c
LDL
eGFR
total_cholesterol
glucose_fasting
creatinine
TSH
potassium
hemoglobin
sodium
HDL
ALT
WBC
AST
troponin_I
BNP
100.0%
100.0%
93.9%
90.8%
80.1%
61.7%
21.7%
0.3%
0.3%
0.3%
0.3%
0.3%
0.3%
0.3%
0.3%
0.1%
0.1%
Mean: 32.4%
0
20
40
60
80
100
% of Results Flagged Abnormal


Primary ICD-10 Code Frequency
Hypertension
Obesity
Hyperlipidemia
Chronic Kidney Disease
Anxiety
Type2 Diabetes
Osteoarthritis
Depression
Coronary Artery Disease
Asthma
Copd
Hypothyroidism
Heart Failure
Atrial Fibrillation
Type1 Diabetes
I10
E66.9
E78.5
N18.3
F41.9
E11.9
M19.90
F32.9
I25.10
J45.909
J44.1
E03.9
I50.9
I48.91
E10.9
0k
10k
20k
30k
40k
50k
60k
Visit count

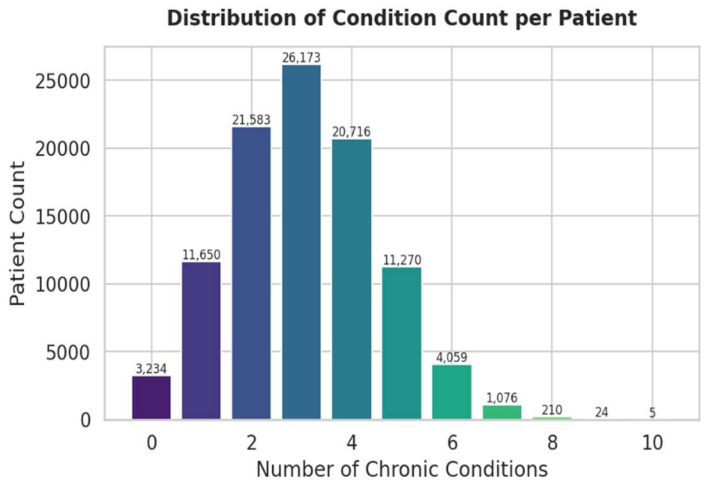


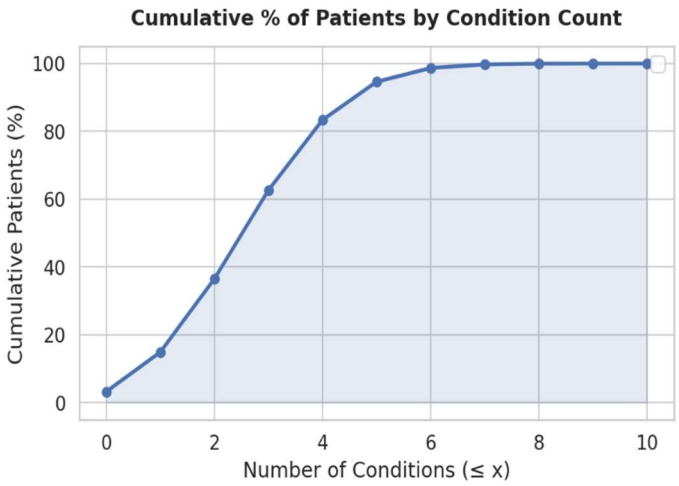
Condition Co-occurrence Heatmap

	HTN	Obesity	Hyperlipid.	CKD	T2DM	OA	Depression	CAD	Anxiety	Asthma	COPD
HTN	0	32,466	28,379	14,041	8,334	7,974	5,353	5,258	4,844	4,762	4,512
Obesity	32,466	0	24,281	12,079	7,172	6,864	4,637	4,334	4,058	4,098	3,693
Hyperlipid.	28,379	24,281	0	10,557	6,172	6,029	3,942	3,950	3,460	3,506	3,285
CKD	14,041	12,079	10,557	0	2,980	3,011	2,018	1,804	1,778	1,677	1,474
T2DM	8,334	7,172	6,172	2,980	0	1,746	1,151	1,152	995	1,127	955
OA	7,974	6,864	6,029	3,011	1,746	0	1,133	1,140	983	1,015	1,011
Depression	5,353	4,637	3,942	2,018	1,151	1,133	0	724	689	681	651
CAD	5,258	4,334	3,950	1,804	1,152	1,140	724	0	666	643	608
Anxiety	4,844	4,058	3,460	1,778	995	983	689	666	0	589	560
Asthma	4,762	4,098	3,506	1,677	1,127	1,015	681	643	589	0	535
COPD	4,512	3,693	3,285	1,474	955	1,011	651	608	560	535	0

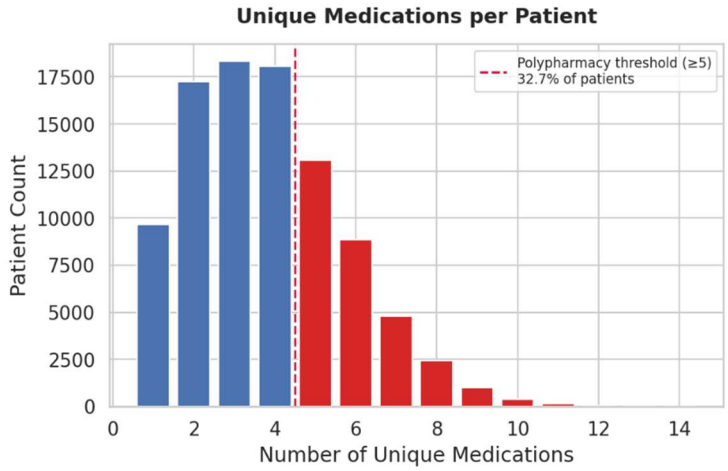
30000
25000
20000
15000
10000
5000
0

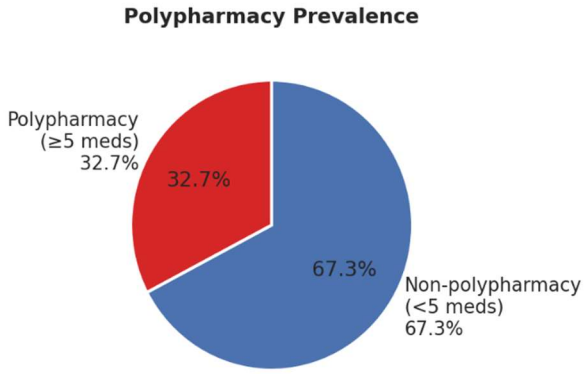
Multimorbidity (Polypharmacy Proxy)



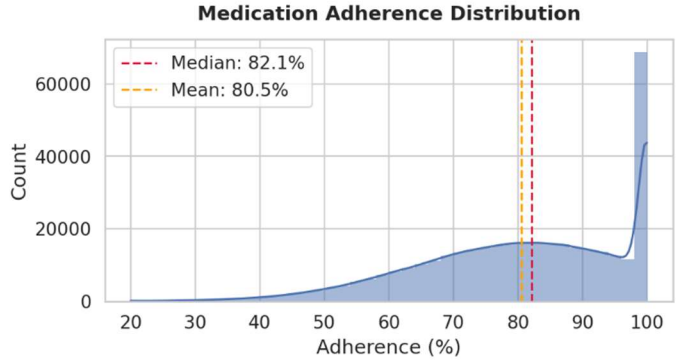


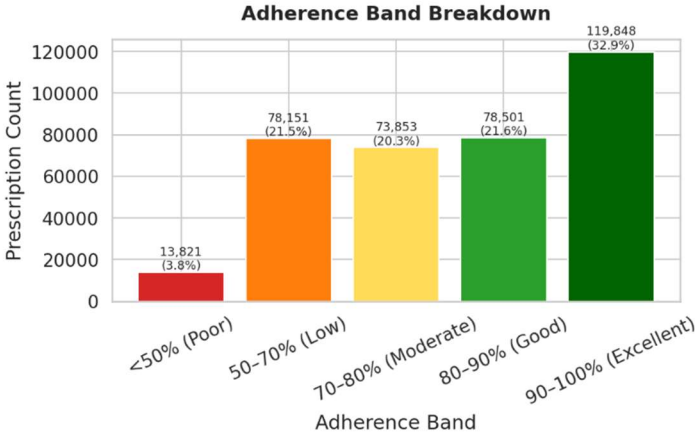
Polypharmacy Analysis (Clinical Threshold ≥5 Medications)

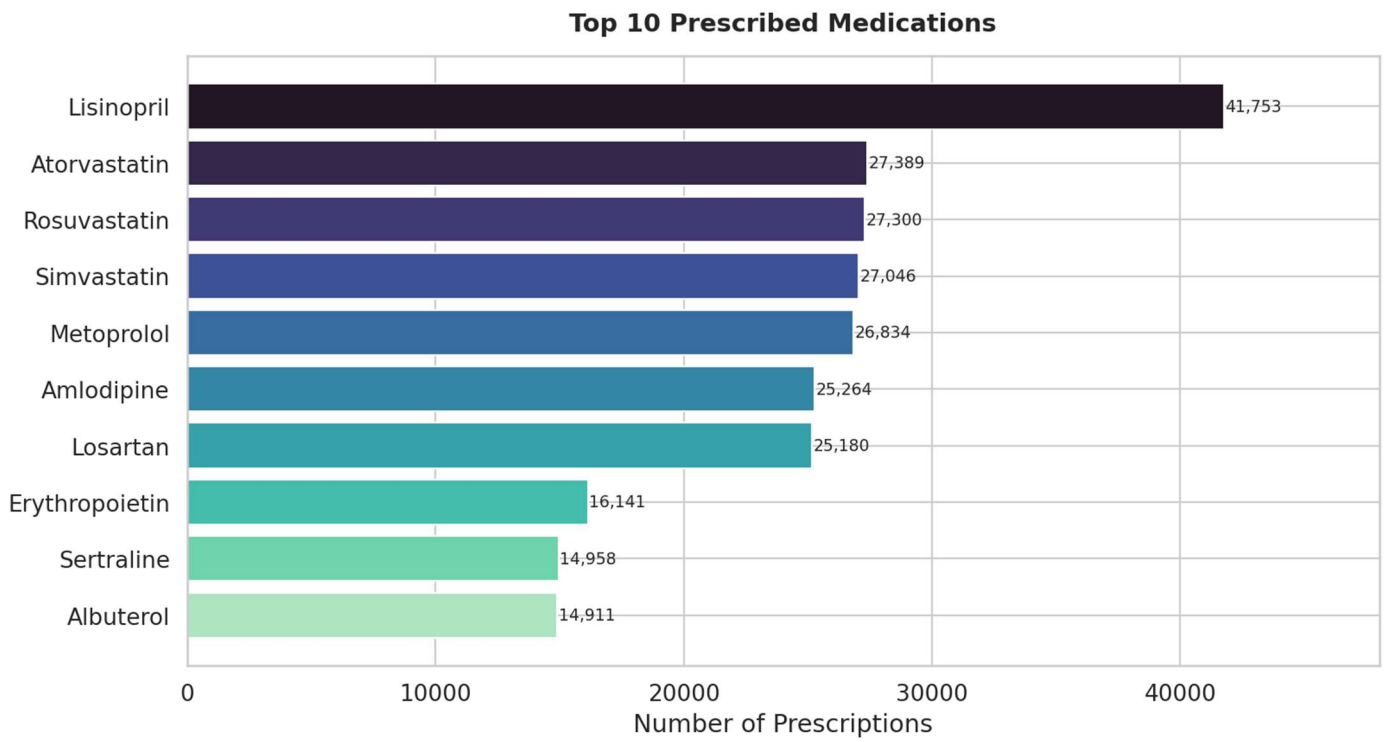


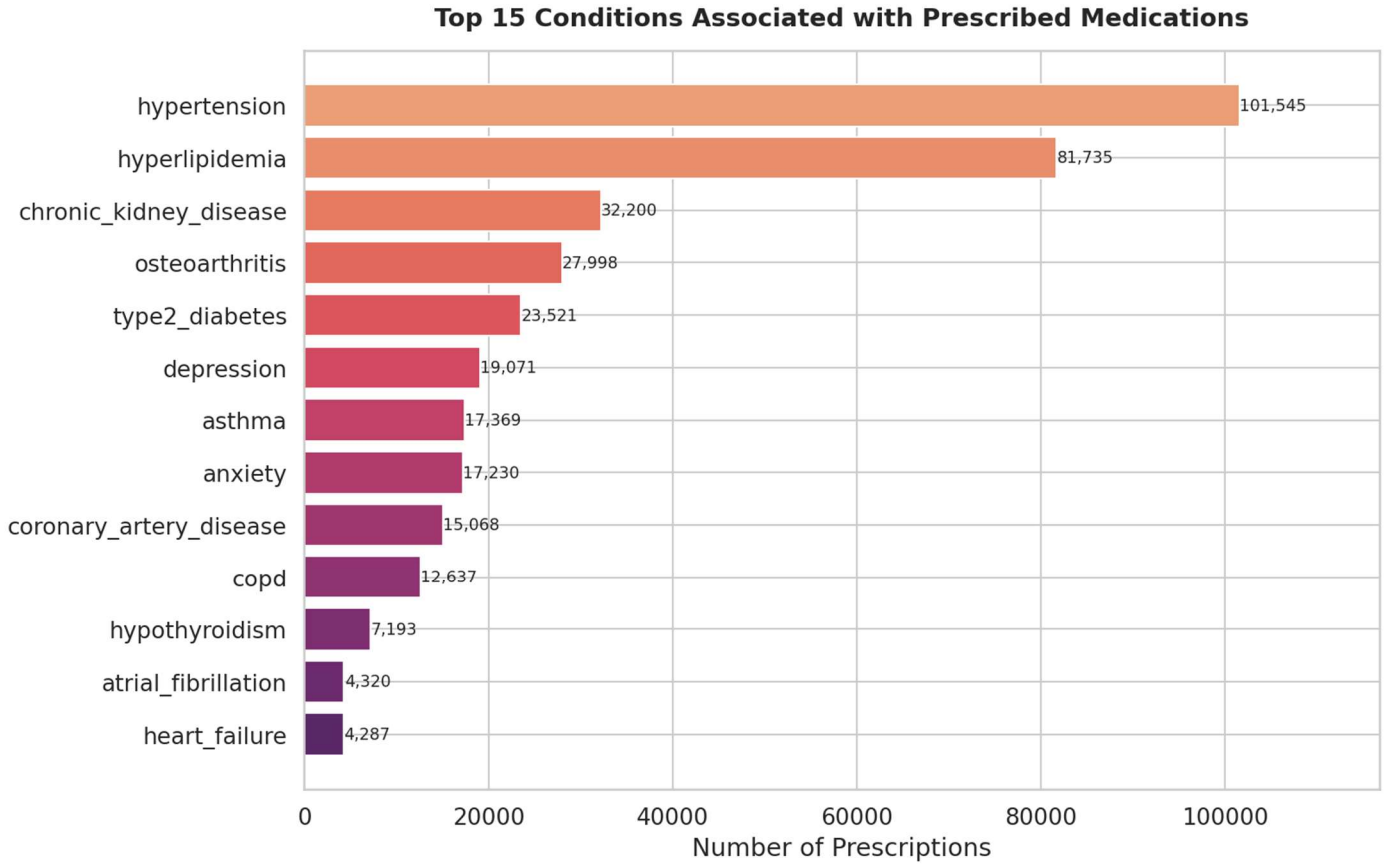


Medication Adherence Analysis






Top 10 Prescribed Medications
Lisinopril
Atorvastatin
Rosuvastatin
Simvastatin
Metoprolol
Amlodipine
Losartan
Erythropoietin
Sertraline
Albuterol
41,753
27,389
27,300
27,046
26,834
25,264
25,180
16,141
14,958
14,911
0
10000
20000
30000
40000
Number of Prescriptions


Top 15 Conditions Associated with Prescribed Medications
hypertension
hyperlipidemia
chronic_kidney_disease
osteoarthritis
type2_diabetes
depression
asthma
anxiety
coronary_artery_disease
copd
hypothyroidism
atrial_fibrillation
heart_failure
101,545
81,735
32,200
27,998
23,521
19,071
17,369
17,230
15,068
12,637
7,193
4,320
4,287
0
20000
40000
60000
80000
100000
Number of Prescriptions